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Data Structure (Version 2019.01.25)

1 Overview

Miscellaneous trial information.

1.1 date

Date of trial data collection (saved in numeric yyyymmddHHMMSS format).

1.2 motionTracker

Which motion tracker was used for this trial (legacy from when we used wiimotes; if blank, assume phasespace was used).

1.3 parameterFile

File path to most recent .prm file loaded before this trial.

1.4 rigCodeVersion

Rig code executable version used to collect the data for this trial.

1.5 singleTrialTag

Trial was tagged (1) or not (0).

1.6 subjectName

Subject for this trial.

1.7 tag

Experiment block this trial was in.

1.8 translationDate

Date this trial was translated (saved in numeric yyyymmddHHMMSS format).

1.9 trialName

State table name for this trial.

1.10 trialNumber

Trial number for this trial (saved as 'Trial[number]').

1.11 trialStatus

Status of this trial.

- 0 = Failed trial
- 1 = Successful trial
- 2 = Error occurred during trial

2 Definitions

Definitions of all possible parameter values for this trial.

2.1 algTargTypeDef

All data sources for analog targets.

2.2 AllPossibleAnalogTargets

Properties of all defined analog targets. Analog targets use voltage data (see Section 4.1 for more information) to determine success.

Note: Each of the following fields contains an array, where each vector element (or matrix row) corresponds to one analog target.

2.2.1 dataSource

Enum value associated with data source (zero-indexed, listed in order in Section 2.1) for each analog target.

2.2.2 dataSourceType

Data source (defined in Section 2.1) for each analog target.

2.2.3 linkedTargets

Linked marker target for each analog target (only defined if dataSourceType is Gaze).

2.2.4 names

Name of each analog target.

2.2.5 tolerances

Number of time points analog window conditions can be not met before window conditions are determined to be not held for each analog target.

2.2.6 windows

Analog target tolerance windows.

For TTL and Touch analog targets, the first value is a voltage minimum threshold.

Gaze tolerance windows match the marker target tolerance window (see Section 5) of its linked target.

2.3 allPossibleIntertrialIntervals

List of all possible intertrial intervals (ITIs) (in ”ms”).

2.4 AllPossibleIntervals

List of all defined interval variables.

Note: Each of the following fields contains an array, where each vector element (or matrix row) corresponds to one interval.

2.4.1 modes

Value selected from list (0) or calculated from formula (1) (see Section 2.21).

2.4.2 names

Name of interval variable.

2.4.3 primary

Whether each interval can be used in state table steps (1) or is only valid for intermediate calculation (0).

2.4.4 values

Listed interval values. Only nonzero values are valid.

2.5 AllPossibleMarkers

Properties of all defined markers.

Note: Each of the following fields contains an array, where each vector element (or matrix row) corresponds to one marker.

2.5.1 channelNumber

Channel number for each marker (only valid for Phasespace markers).

2.5.2 dataSource

Enum value associated with data source for each marker

- 0 = Phasespace/Wiimotes
- 1 = Neural Decoded
- 2 = Auto-monkey
- 3 = Error
- 4 = Force

2.5.3 dataSourceType

Data source (Phasespace/Wiimotes, Neural Decoded, Auto-monkey, Error, Force) for each marker.

2.5.4 names

Name of each marker.

2.6 AllPossibleStateTables

Properties of all possible state tables.

Note: Each of the following fields contains an array, where each vector element (or matrix row) corresponds to one state table.

2.6.1 globalTimeout

Global timeout value for each state table (in ”ms”). If global timeout is applied, then when the specified time has elapsed, the trial will jump to the state table step defined in globalTimeoutFailureStep.

2.6.2 globalTimeoutFailureStep

Step to jump to once global timeout period has elapsed for each state table.

2.6.3 names

Name of each state table.

2.6.4 probabilities

Probability for each state table (in percentage).

2.7 AllPossibleTargets

Properties of all possible marker targets.

Note: Each of the following fields contains an array, where each vector element (or matrix row) corresponds to one target.

2.7.1 formulas

Formula for calculating location of each marker target (valid only for Cartesian Formula mode marker targets).

2.7.2 modes

Mode of marker target location selection (see Section 2.26) for each marker target.

2.7.3 names

Name of each marker target.

2.7.4 primary

Whether each marker target can be referenced in state table steps (1) or is only valid for intermediate calculation (0).

2.7.5 selectionModes

Mode of selecting target locations from list (see Section 2.27).

2.7.6 shape

Shape of marker target window (see Section 2.30).

2.7.7 window

Tolerance window for each marker target (in mm).

For box-shaped (see Section 2.30) tolerance windows, these values represent x, y, and z tolerance values. For cylindrical tolerance windows, the numerical values represent radius and depth (the second value is NaN). For spherical tolerance windows, the numerical value represents radius (the second and third values are NaNs).

2.7.8 xValues

Lists of possible x position values (in mm).

2.7.9 yValues

Lists of possible y position values (in mm).

2.7.10 zValues

Lists of possible z position values (in mm).

2.8 AllPossibleTubeFilepaths

[1] File path to each tube.

2.9 AllPossibleTubeGroups

Group(s) assigned to each tube. Zero values are only valid if listed first.

2.10 AllPossibleTubeLengths

Default distance between endpoints of each tube.

2.11 AllPossibleTubeNames

Name of each tube.

2.12 AllPossibleTubePathLengths

Default path length for each tube

2.13 AllPossibleTubeRenderTypes

Graphical rendering type for each tube.

2.14 allPossibleVibrotactorDefinitions

Types of motors.

2.15 allPossibleVibrotactorDimensions

Possible directional modes of motor activation.

[1] *Consider consolidating tube properties into struct.*

2.16 allPossibleWaveformShapes

Types of tactor or pager motor waveform shapes.

2.17 analogChannelNames

Types of information saved in analogData (see Section 4.1) corresponding to each column.

2.18 autoMonkeyProfileDef

Types of auto-monkey trajectory profiles:

- Linear: Linear speed profile
- Bell-shaped: Bell-shaped speed profile
- Trajectory: User-defined times and positions

2.19 gazeCheckDef

Types of checks on eye position.

2.20 handCheckDef

Types of checks on marker targets (and analog targets of types other than TTL).

2.21 intervalModeDef

Modes of selecting or calculating interval values.

2.22 MarkerSpeedProfiles

Properties of all possible auto-monkey trajectory profiles.

Note: Each of the following fields contains an array, where each vector element (or matrix row) corresponds to one auto-monkey profile.

2.22.1 finalValues

Final cursor speed value (in mm/s if absolute speed) for each auto-monkey trajectory profile.

2.22.2 initialValues

Initial cursor speed value (in mm/s if absolute speed) for each auto-monkey trajectory profile.

2.22.3 names

Name of each auto-monkey trajectory profile

2.22.4 relativeSpeed

Whether speed is defined relative to start and end locations (1) or in absolute magnitude (0).

2.22.5 types

Type of trajectory for each auto-monkey trajectory profile (see Section 2.18).

2.23 motorModeDef

Activation modes for tactors or pager motors.

2.24 rawDecodeColumnNames

Types of information saved in rawDecode (see Section 4.2.4) corresponding to each column.

2.25 snippetInfoRowNames

Types of extracted TDT snippet information corresponding to each row.

2.26 targetModeDef

Modes of selecting or calculating target locations.

2.27 targetSelectionModeDef

Modes of selecting target locations from possible values (valid only for Cartesian List mode targets).

2.28 targetShapeDef

Graphical display shape options for marker targets.

2.29 targetTypeDef

Types of graphical display objects.

2.30 targetWindowShapeDef

Types of marker target tolerance window shapes.

2.31 ttlCheckDef

Types of checks on TTL voltage.

2.32 tubeCheckDef

Types of checks on tube window.

2.33 vrPhasespaceTransformation

Parameters for calculating VR space position from Phasespace coordinates.

2.33.1 rotation

Rotation about x, y, and z axes (in degrees).

2.33.2 scaling

Scaling of x, y, and z axes.

2.33.3 translation

Translation along x, y, and z axes (in mm).

3 Parameters

3.1 eyeTrackerCalibrationMatrix

Matrix mapping eye tracker analog voltage outputs to Phasespace coordinates. The first column represents x and y translation (in mm). The next two columns are gain multipliers for the left eye x and y voltage outputs (in mm/V). If binocular tracking was used, the last two columns are gain multipliers for the right eye x and y voltage outputs (in mm/V).

3.2 ForceParameters

CST error cursor parameters. [2]

3.2.1 initialLambda

Initial lambda value for the dynamical system.

3.2.2 inputMethod

Data source used as input to dynamical system

- 0 = Force (i.e., pinch force)
- 1 = Decoded Cursor
- 2 = Phasepace

3.2.3 kicms

Intracortical microstimulation gain (no ICMS if 0).

3.2.4 kv

Visual gain (no visual feedback if 0).

3.2.5 kvib

Vibrotactile gain (no tactor feedback if 0).

3.2.6 refValue

Reference (target) value of the dynamical system. This value is subtracted from the raw input before input is passed through the dynamical system.

3.2.7 slowfastRate

Rates defining how lambda changes based on the error. The fast rate is used until the error surpasses a given threshold, at which point the slow rate is used.

3.2.8 systemName

Description of the dynamical system.

[2] *Neither error threshold nor visual direction are saved to data structure. Additionally, kvib does not appear to be used online. If kv is 0, error cursor input appears to always be 0. In old vibrotactile CST data, is kv 1 but error cursor not displayed? Or was this written out in the dynamical system response code?*

3.2.9 systemOrder

Order of the dynamical system (1 or 2).

3.2.10 visCursorOffsets

Offset applied to cursor for visual feedback (in mm).

3.3 LockZParameters

Parameters for overwriting Phasespace z coordinate.

3.3.1 lockZ

Overwrite Phasespace z coordinate (1) or not (0).

3.3.2 lockZCoordinate

Value to use in place of current Phasespace z coordinate (in mm).

3.4 markerJitter

Phasespace jitter parameters.

3.4.1 A

Gain multiplier for previous jitter value.

3.4.2 controlJitter

Use jittered cursor position for online task control (1) or not (0).

3.4.3 S

Standard deviation of jitter (in mm).

3.4.4 visibleJitter

Display cursor visually with jitter (1) or not (0).

3.5 MotorParameters

Properties of tactor and pager motor activation.

Note: Each of the following fields contains an array, where each vector element (or matrix row) corresponds to one motor.

3.5.1 amplitudeRanges

Minimum and maximum amplitudes for each motor for this trial (in V).

3.5.2 carrierFrequencyRanges

Minimum and maximum carrier frequencies for each motor (in Hz).

3.5.3 distanceRanges

Minimum and maximum distances for each motor (in mm).

3.5.4 dutyCycleRanges

Minimum and maximum duty cycle values for each motor (in percentage).

3.5.5 forceToZeroIfOutsideRange

Coerce value to 0 if outside range (1) or not (0) for each motor.

3.5.6 ignoreIfOutsideRange

Ignore value if outside range (1) or not (0) for each motor.

3.5.7 modulationFrequencyRanges

Minimum and maximum modulation frequencies for each motor (in Hz).

3.5.8 vibrotactorDimensions

Dimension for each motor (see Section 2.15).

3.5.9 vibrotactorsNumber

Tactor number for each motor.

3.5.10 vibrotactorTypes

Type (see Section 2.14) of each motor.

3.5.11 waveformShapes

Waveform shape (see Section 2.16) for each motor.

3.6 Pseudorandomness

3.6.1 pseudorandom

Pseudorandomize target presentation for this trial type (1) or not (0).

3.6.2 pseudorandomReset

Pseudorandomness was reset this trial (1) or not (0). [3]

3.7 StateTable

Each element of this array represents the trial progression parameters for a single state of this trial. These array elements contain the following fields:

[3] *Pseudorandomness reset value is always reset to 0 before being sent back to Host.*

3.7.1 Analog

Properties of analog targets referenced in this state.

Note: Each of the following fields contains an array, where each vector element (or matrix row) corresponds to one analog target.

3.7.1.A. dataSource
Data source for each analog target.

3.7.1.B. linkedTarget
Name of linked target for each analog target.

3.7.1.C. linkedTargetIndex
Index of linked target, if valid, in TrialTargets array (see Section 3.9).

3.7.1.D. targetName
Name of analog target.

3.7.1.E. targetNameIndex
Index of analog target in AllPossibleAnalogTargets array (see Section 2.2).

3.7.1.F. targetWindowCheckType
Enum value of analog target window check type (see Sections 2.20 and 2.31).

3.7.1.G. window
Tolerance window for each analog target (see Section 2.2.6).

3.7.2 applyGlobalTimeout

Apply global timeout (1) or not (0) (see Section 2.6.1).

3.7.3 autoMonkeySpeedProfileName

Auto-monkey trajectory profile for this state (N/A if no auto-monkey trajectory profile selected).

3.7.4 Display

Graphical display properties for this state.

3.7.4.A. cursor
Display cursor (1) or not (0) for this state.

3.7.4.B. endTarget
Display endTarget (1) or not (0) for this state.

3.7.4.C. markers
Marker numbers to display and whether to freeze marker (1) or not (0) for this state.

3.7.4.D. startTarget
Display startTarget (1) or not (0) for this state.

3.7.4.E. tube
Display tube (1) or not (0) for this state.

3.7.5 Gaze

Gaze fixation target for this state.

3.7.5.A. targetName
Name of linked gaze target.

3.7.5.B. targetNameIndex
Index of linked target, if valid, in TrialTargets array.

3.7.5.C. targetWindowCheckType
Enum value of gaze target window check type (see Section 2.20).

3.7.5.D. window
Tolerance window for the linked gaze target (see Section 5).

3.7.6 Hand

Marker target properties for this state.

Note: Each of the following fields contains an array, where each vector element (or matrix row) corresponds to one marker target.

3.7.6.A. targetMarker
Name of marker used for window checks for this target.

3.7.6.B. targetName
Name of marker target.

3.7.6.C. targetNameIndex
Index of marker target in TrialTargets array.

3.7.6.D. targetWindowCheckType
Enum value of marker target window check type (see Section 2.20).

3.7.6.E. window
Tolerance window for each marker target.

Column format:

- x coordinate of target center (in mm)
- y coordinate of target center (in mm)
- z coordinate of target center (in mm)
- x window length (see Section 2.7.7) (in mm)
- y window length (in mm)
- z window length (in mm)

3.7.7 Interval

Interval used for this state.

3.7.7.A. length
Value of interval (in ”ms”).

3.7.7.B. name
Name of interval.

3.7.8 Motor

Motor control parameters for this state.

3.7.8.A. mode
Mode of motor activation (see Section 2.23).

3.7.8.B. target
Location of motor target (x, y, z coordinates, in mm).

Note: If no motor target defined, location will be reported as [-1 -1 -1].

3.7.8.C. targetName
Name of motor target.

3.7.8.D. targetNameIndex
Index of motor target in TrialTargets array.

3.7.9 passTable

Define the trial as successful if state is reached (1) or not (0).

3.7.10 resetGlobalTimeout

Reset global timeout counter (1) or not (0) (see Section 2.6.1) for this state.

3.7.11 stateName

Name of this state.

3.7.12 StateTargets

Displayed target properties for this state.

Note: Each of the following fields contains an array, where each vector element (or matrix row) corresponds to one displayed target.

3.7.12.A. color
Color of displayed target.

3.7.12.B. location
Location of displayed target (x, y, z coordinates, in mm).

3.7.12.C. nameIndex
Index of displayed target in TrialTargets array.

3.7.12.D. names
Name of displayed target.

3.7.12.E. shape
Shape of displayed target (see Section 2.28).

3.7.12.F. size
Size of displayed target (in mm).

3.7.12.G. type
Graphical target type (see Section 2.29) of displayed target.

3.7.13 togglePhotodiodeTarget

Toggle photodiode target (1) or not (0) for this state.

3.7.14 TTL

TTL output check type and output pattern for this state. [4]

[4] *Output pattern names and values not saved to data structure.*

3.7.15 tubeColor

Color of tube for this state (only valid if tube is displayed).

3.7.16 tubeMarker

Marker for tube control for this state (only valid if tube window is checked).

3.7.17 tubeSection

Section of tube for control and display for this state (if 0, the whole tube is used for control and display).

3.7.18 tubeTexture

Texturing applied to tube for this state.

3.7.19 tubeWindow

Tube tolerance radius and depth for this state (in mm).

3.7.20 tubeWindowCheckType

Tube window check type for this state (see Section 2.32).

3.7.21 useTubeTexture

Use tube texture (1) or not (0) for this state.

3.7.22 windowFailState

State to jump to if this state fails (0 represents trial end).

3.7.23 windowPassState

State(s) to jump to if this state succeeds (0 represents trial end). [5]

3.8 stateTableText

All state table cell values for this trial.

3.9 TrialTargets

Properties of all displayed targets for this trial.

Note: Each of the following fields contains an array, where each vector element (or matrix row) corresponds to one displayed target.

3.9.0.A. color
Color of each target.

3.9.0.B. names
Name of each target.

3.9.0.C. size
Size of each target.

[5] *State table target logic (AND vs OR) not saved in data structure.*

3.9.0.D. statesUsed
States where each target was used.

3.9.0.E. window
Tolerance window of each displayed target (see Section 5).

3.10 TrialTubeParameters

3.10.0.A. displayRadii
Graphical radius of displayed tube (in mm).

3.10.0.B. numberSections
Total number of sections of tube.

3.10.0.C. pseudorandomizeTubes
Pseudorandomize tube presentation (1) or not (0).

3.10.0.D. pseudorandomTubeTarget
Name of target linked to tube pseudorandomness.

3.10.0.E. rotation
Rotation of target trajectory (in degrees).

3.10.0.F. size
Size of tube (valid only for very old data).

3.10.0.G. trajectory
Downsampled tube trajectory coordinates (in mm).

3.10.0.H. tubeName
Name of tube.

3.10.0.I. tubeType
Type of tube (valid only for very old data).

3.10.0.J. window
Radius and depth of tube control window (in mm).

3.11 visualLag

Visual lag applied to phasespace data (in ”ms”).

3.12 visualLagRandom

Randomize visual lag (1) or not (0).

4 TrialData

Data used for task control for this trial.

4.1 analogData

Voltage data from analog input channels. Each row represents one frame.

The ”actual time” column collected in newer executables contains the number of 500 us periods elapsed from the end of the last frame. Thus the actual time in ms can be calculated by multiplying this column by 0.5.

4.2 Decoder

4.2.1 averagedSpikeBinHistory

Average spike count history for this trial.

4.2.2 decoderType

Type of decoder (e.g. Kalman (for data collected with older executables), LDS, GPFA) for this trial.

4.2.3 Parameters

Parameters of decoder for this trial.

4.2.4 rawDecode

Raw decode information for this trial. Each row represents one decoder prediction. Column information can be found in Section 2.24.

4.2.5 rawSpikeBins

Raw spike bin information for this trial. Raw spike bins used online for decoder prediction.

4.2.6 runDecoder

Run decoder (1) or not (0) for this trial.

4.3 intertrialInterval

Intertrial interval for this trial (in ”ms”).

4.4 Marker

4.4.1 autoMonkey

Auto-monkey cursor data for this trial. The auto-monkey follows a user-defined trajectory.

4.4.1.A. position
Auto-monkey cursor positions (in mm).

4.4.1.B. time
Auto-monkey cursor time (in ”ms”).

4.4.2 errorCursor

CST error cursor data for this trial. The error cursor is defined by a dynamical system.

Column format:

- Error cursor x coordinate (in mm)
- Error cursor y coordinate (in mm)
- Error cursor z coordinate (in mm)
- Time (in "ms")
- Lambda value (in rad/s)
- Dynamical system input value (in mm)

4.4.3 frequency

Phasespace frequency for this trial.

4.4.4 rawPositions

Phasespace position data for this trial.

Column format:

- Phasespace marker ID
- Phasespace raw x coordinate (in mm)
- Phasespace raw y coordinate (in mm)
- Phasespace raw z coordinate (in mm)
- Phasespace frame number
- Phasespace synchronized raw time (in ms, calculated from frame number)
- Phasespace synchronized raw time + visual lag (in ms)
- Phasespace x jitter (in mm)
- Phasespace y jitter (in mm)
- Phasespace z jitter (in mm)
- Phasespace jittered x coordinate (in mm)
- Phasespace jittered y coordinate (in mm)
- Phasespace jittered z coordinate (in mm)

4.4.5 rotation

Phasespace rotation value (in degrees).

4.4.6 rotationCenter

Coordinates of phasespace rotation origin (in mm).

4.4.7 scaling

Scaling of phasespace data.

4.4.8 SyncParameters

4.4.8.A. phasespaceFrame
Phasespace frame number value at start of sync pulse.

4.4.8.B. pulseLength
Length of sync pulse from start time (in ”ms”).

4.4.8.C. pulseLogic
Digital line output value of sync pulse (1 = 5V, 0 = 0V). Outside of sync pulse, digital output line is set to opposite binary value.

4.4.8.D. startTime
Start time of sync pulse (in ”ms”).

4.4.9 trackingMarker

Phasespace marker number used for movement tracking.

4.5 numberOfManualRewards

Number of manual rewards given via pushbutton.

4.6 stateTransitions

State transition data. The first row is state number. The second row is the frame number at which this state began. The last state (-1) represents the end of the trial.

4.7 TDT

Neural data saved for this trial.

4.7.1 bb_samplingRate

Sampling rate for broadband waveform for this trial (in Hz).

4.7.2 bb_waveforms

Broadband waveform data for this trial (in V).

4.7.3 Block

Name of original block in TDT tank where neural data was saved for this trial.

4.7.4 brainArea

Brain area where this trial's subject's array was implanted.

4.7.5 lfpCutoffFrequencies

Local field potential (LFP) filter frequencies for this trial (in Hz).

4.7.6 lfpCutoffFrequenciesFs

Sampling rate for LFP filter frequencies for this trial (in Hz).

4.7.7 lfpFs

Sampling rate for LFP data for this trial (in Hz).

4.7.8 lfpWaveforms

LFP waveform data for this trial (in V).

4.7.9 numberOfChannels

Number of channels input to the PZ2 preamplifier for this trial.

4.7.10 rawTDT

Vector of UDP packets of neural data from the RZ2 for this trial.

Each packet contains a header, RT time, TDT sample number, and 32-bit words representing spike counts of multiple channels.

4.7.11 snippetInfo

Selected snippet event information for this trial (see Section 2.25).

4.7.12 snippetInfoCutoffFrequencies

Neural filter cutoff frequencies for this trial (in Hz).

4.7.13 snippetInfoCutoffFrequenciesFs

Sampling rate for neural filter cutoff frequencies for this trial (in Hz).

4.7.14 snippetWaveforms

Snippet waveform data for this trial (in V).

4.7.15 sortName

Name of sort data for this trial.

4.7.16 tankPath

Path to tank directory for neural data stored for this trial.

4.7.17 threshold

Threshold values for this trial (in V).

4.7.18 thresholdFs

Sampling rate for threshold values for this trial (in V).

4.7.19 thresholdStatic

Manual (1) or automatic (0) threshold values for this trial.

4.7.20 thresholdStaticFs

Sampling rate of manual threshold status for this trial (in Hz).

4.7.21 timestampUnits

Neural event timestamp units for this trial. Millisecond (ms) or microsecond (us) precision can be used.

4.7.22 trialID

Trial ID value sent to TDT to mark beginning of this trial.